Empirical Note: A Tiny (≈50-Line) TSP Heuristic Showing ~99% Local-Move Rejection on TSPLIB Instances

Abstract
We present a compact (≈50-line) constructive + local-search baseline ("V2") for the symmetric Traveling Salesman Problem (TSP), designed for practical, budget-tunable operation rather than academic competitiveness. We evaluate "model V2" on six TSPLIB instances (berlin52, eil51, pr76, eil101, kroA100, kroA150), compute tour costs under TSPLIB distance rules (e.g., EUC_2D integer rounding), and report gaps relative to the canonical Concorde TSPLIB optimal tour lengths.

Despite its tiny footprint and no instance-specific tuning, V2 exhibits a distinctive behavioral signature: extreme local-move pruning, with 2-opt acceptance ≈0.02–0.38% and Or-opt ≈0.015–0.054% across all instances (i.e., >99.6–99.98% of evaluated moves are rejected by the accept rule). V2 also exposes simple budget knobs (restarts; number of 2-opt / Or-opt passes) that move the solver along a quality-vs-budget frontier while preserving this pruning behavior. We discuss V2 as a practical routing primitive for small-to-medium problem sizes and embedded settings, and note that more advanced models (including non-Euclidean/3D variants) exist but are out of scope for this note.

1. Introduction

Many real-time robotic and navigation systems must repeatedly solve TSP-like subproblems under tight compute budgets and on off-the-shelf hardware. Our objective was to build an extremely small routing kernel that is predictable, tunable, and fast enough to be practical in such environments. Rather than pursue state-of-the-art accuracy, we asked: *how far can a tiny constructive + local-search baseline be pushed if pruning efficiency and budget control are prioritized?*

All experiments used instances from the TSPLIB95 benchmark library. Tour costs were evaluated exactly according to the distance specifications provided for each instance (e.g., EUC_2D integer rounding), and performance was measured as the percentage gap relative to the published optimal tour lengths from the Concorde solver, the standard reference for exact TSPLIB solutions.

(This note evaluates only the 2D Euclidean baseline; no non-Euclidean or 3D variants are included.)

2. Method (V2 Baseline)

Solver footprint. V2 is an intentionally small (≈50-line) constructive + local-search heuristic. It first builds a tour using a sector-wise greedy rule, then applies a set of 2-opt and Or-opt sweeps. No instance-specific tuning was performed; the same settings were used for all instances.

Tunability. V2 exposes simple budget knobs—such as the number of restarts and the number of 2-opt / Or-opt passes—that allow compute budget and quality to be traded off in a predictable manner.

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Evaluation protocol.

– Instances: berlin52, eil51, pr76, eil101, kroA100, kroA150.
– Distance rules: TSPLIB-specified metric for each instance (e.g., EUC_2D integer rounding)
– Reference optima: Concorde TSPLIB optimal tour lengths.
– Gap definition: gap=(V2 length−opt).

Instrumentation. To quantify pruning behavior, we record:

- 2-opt: checks → applies
- Or-opt: attempts → applies
- Total greedy nearest-neighbor comparisons

3. Results

3.1 TSPLIB accuracy (V2 single-tour lengths; gaps vs Concorde)

Instance	V2 length	Concorde optimal	Gap
berlin52	7,711	7,542	**+2.24%**
eil51	437	426	**+2.58%**
pr76	133,827	108,159	**+23.75%**
eil101	709	629	**+12.72%**
kroA100	39,492	21,282	**+85.60%**
kroA150	60,599	26,524	**+128.53%**

Reference optima from the canonical Concorde TSPLIB table. Instances from TSPLIB95; tour costs computed under the suite's distance rules (e.g., EUC_2D integer rounding).

Interpretation. The V2 baseline is untuned and very lean, so we do not expect strong quality on larger instances. The small-instance gaps (e.g., berlin52, eil51) are reasonable; larger instances show wider gaps. The central observation, however, is V2's behavioral stability and tunability rather than peak quality.

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3.2 Local-move pruning (accept-rates)

Instance	Greedy comps	2-opt checks → applies	2-opt accept-rate	Or-opt attempts → applies	Or-opt accept-rate
berlin52	144	65,659 → 153	**0.233%** (→ **99.767% rejected**)	13,480 → 5	**0.037%** (→ **99.963%**)
eil51	138	32,775 → 125	**0.381%** (→ **99.619%**)	9,886 → 5	**0.050%** (→ **99.950%**)
pr76	324	199,773 → 200	**0.100%** (→ **99.900%**)	56,332 → 20	**0.036%** (→ **99.964%**)
eil101	588	403,533 → 200	**0.0496%** (→ **99.950%**)	131,441 → 20	**0.0152%** (→ **99.985%**)
kroA100	576	175,129 → 200	**0.114%** (→ **99.886%**)	37,173 → 20	**0.0538%** (→ **99.946%**)
kroA150	1,332	329,572 → 200	**0.0607%** (→ **99.939%**)	105,039 → 20	**0.0190% (→ 99.981% rejected)**

Observation. Across all six instances, V2 accepts only 0.02–0.38% of 2-opt moves and 0.015–0.054% of Or-opt moves, corresponding to >99.6–99.98% rejection. This consistently high selectivity is a distinctive characteristic of the baseline.

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4. Discussion

- **Practicality and predictability.** V2 is designed as a small, transparent routing primitive suitable for low-power or real-time settings. Its most notable property is its consistent local-move selectivity, with acceptance rates well below 1% across all instances.
- **Accuracy vs scale**. On small instances, gaps are modest; on larger instances (100–150 nodes) the untuned baseline exhibits wider gaps, which we consider acceptable for this note's goal (demonstrating behavioral stability and tunability).
- **Relation to classical heuristics**. Algorithms such as LKH/LKH-3 achieve strong quality using smart neighborhood restrictions (candidate sets, don't-look bits) to reduce move generation. V2 shows a complementary property: aggressive move rejection while keeping the codebase minimal and the knobs simple.
- **Scope**. Many navigation workloads involve ≤ 60 locations, where tunability and predictable behavior at a chosen budget can be more valuable than asymptotic competitiveness.
- **Future work**. The V2 baseline served as a scaffold for more advanced solvers, including non-Euclidean / 3D variants with similar pruning characteristics; these are proprietary and out of scope here.

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5. Reproducibility

- **Evaluation protocol:**
  – Instances from TSPLIB (e.g., berlin52, eil51, pr76, eil101, kroA100, kroA150).
  – Costs computed under TSPLIB distance rules (e.g., EUC_2D integer rounding).
  – Gaps vs Concorde TSPLIB optimal tour lengths.
- **Tour-length verifier (optional):** We provide a minimal EUC_2D verifier that recomputes the cost of any given permutation under TSPLIB rules; this reveals nothing about the heuristic and can be used to confirm the reported V2 lengths.
- **Independent audit (optional):** We include 1–2 NEOS/Concorde job IDs demonstrating optimality for select instances, providing an external reference point.

Data:

The following permutations are provided for reproducibility. Tour costs may be recomputed directly using the TSPLIB EUC_2D distance rule.

Instance: berlin52
V2_TSPLIB_LENGTH: 7711
V2_TOUR_SEQUENCE: [31, 18, 3, 17, 21, 42, 7, 2, 30, 23, 20, 50, 29, 16, 46, 48, 24, 5, 15, 6, 4, 25, 12, 28, 27, 26, 47, 13, 14, 52, 11, 51, 33, 43, 10, 9, 8, 41, 19, 45, 32, 49, 35, 36, 39, 40, 38, 37, 34, 44, 1, 22]

Instance: eil51
V2_TSPLIB_LENGTH: 133827
V2_TOUR_SEQUENCE: [32, 1, 22, 2, 16, 11, 38, 5, 15, 45, 33, 39, 10, 30, 49, 9, 50, 34, 21, 29, 20, 35, 36, 3, 28, 31, 26, 8, 48, 6, 23, 7, 43, 24, 14, 25, 13, 41, 40, 19, 42, 44, 37, 17, 4, 18, 47, 12, 46, 51, 27]
V2_TSPLIB_LENGTH: 437 Instance: pr76 V2_TOUR_SEQUENCE: [8, 7, 6, 5, 4, 3, 2, 75, 76, 1, 23, 22, 21, 25, 24, 46, 45, 44, 48, 47, 69, 68, 70, 67, 50, 49, 52, 53, 54, 42, 43, 28, 27, 26, 20, 19, 31, 30, 29, 33, 32, 36, 18, 15, 13, 17, 11, 12, 74, 37, 38, 35, 34, 40, 60, 59, 41, 61, 62, 57, 63, 64, 73, 72, 71, 65, 66, 51, 55, 56, 58, 39, 16, 10, 9, 14]

Instance: eil101
V2_TSPLIB_LENGTH: 709
V2_TOUR_SEQUENCE: [60, 83, 45, 8, 46, 47, 36, 49, 64, 11, 19, 48, 82, 18, 52, 7, 88, 31, 62, 10, 63, 90, 32, 70, 30, 20, 66, 65, 71, 35, 9, 51, 33, 81, 34, 78, 79, 3, 77, 76, 50, 1, 69, 27, 101, 53, 28, 26, 12, 80, 68, 29, 24, 54, 55, 25, 39, 67, 23, 56, 4, 72, 74, 75, 22, 41, 15, 43, 38, 44, 14, 42, 57, 2, 58, 40, 21, 73, 95, 99, 5, 84, 17, 86, 16, 61, 59, 94, 13, 87, 37, 91, 85, 100, 93, 98, 92, 97, 96, 6, 89]

Instance: kroA100
V2_TSPLIB_LENGTH: 39492
V2_TOUR_SEQUENCE: [79, 18, 24, 38, 99, 36, 59, 17, 15, 11, 32, 45, 91, 98, 23, 93, 28, 67, 58, 5, 39, 30, 46, 12, 86, 27, 35, 62, 60, 77, 20, 57, 9, 7, 87, 51, 81, 25, 40, 64, 2, 44, 50, 73, 61, 69, 68, 82, 76, 78, 71,

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48, 3, 85, 13, 100, 34, 43, 96, 29, 83, 55, 41, 14, 33, 52, 37, 95, 54, 89, 31, 80, 56, 42, 8, 92, 1, 75, 97, 4, 65, 26, 66, 70, 22, 94, 88, 16, 53, 19, 6, 63, 47, 21, 74, 72, 10, 84, 49, 90]

Instance: kroA150
V2_TSPLIB_LENGTH: 60599
V2_TOUR_SEQUENCE: [137, 79, 134, 53, 106, 90, 49, 6, 63, 113, 10, 84, 18, 24, 38, 104, 111, 102, 99, 36, 127, 72, 21, 74, 59, 141, 17, 15, 11, 32, 45, 98, 91, 109, 47, 131, 93, 28, 123, 3, 100, 30, 121, 39, 33, 126, 82, 146, 64, 44, 13, 5, 101, 132, 120, 115, 12, 77, 62, 9, 87, 145, 150, 149, 83, 140, 43, 41, 52, 37, 76, 40, 54, 69, 25, 125, 51, 55, 27, 46, 48, 96, 147, 116, 144, 133, 105, 142, 130, 1, 92, 8, 148, 67, 108, 58, 61, 7, 34, 112, 71, 68, 73, 2, 95, 103, 128, 136, 14, 107, 85, 20, 60, 86, 23, 110, 78, 29, 35, 57, 117, 135, 81, 50, 114, 138, 89, 122, 80, 118, 124, 26, 66, 129, 65, 143, 42, 97, 56, 119, 31, 75, 139, 19, 4, 70, 94, 22, 16, 88]

References

- **TSPLIB / TSPLIB95** (benchmark suite, instance info, data access).
- **Concorde's TSPLIB optimal tour lengths** (authoritative table).
- **TSPLIB distance rules / FAQ** (EUC_2D integer rounding; GEO, CEIL_2D notes).
- **LKH / LKH-3 resources** (neighborhood restriction, candidate sets, don't-look bits).

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